

**Composition to Verify Inverses**

Algebra II | Unit 1 | Day 13 | TEK A2.2.D

**Objective:** I can compose two functions and use the results  $f(f^{-1}(x)) = x$  and  $f^{-1}(f(x)) = x$  to prove whether they are inverses of each other.

**Vocabulary**

**Composition** — putting one function \_\_\_\_\_ another. Written  $f(g(x))$  or  $(f \circ g)(x)$ .

**Order matters** — work from the \_\_\_\_\_. Evaluate  $g$  first, then feed the result into  $f$ .

**The inverse test** —  $f$  and  $g$  are inverses when \_\_\_\_\_ AND \_\_\_\_\_. Both directions.

**Part 1 — Warm Up on Composition**

Let  $f(x) = 3x - 5$  and  $g(x) = (x + 5)/3$ . Work inside first.

$$g(2) = \underline{\hspace{2cm}} \quad f(4) = \underline{\hspace{2cm}} \quad f(g(4)) = f(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$$

$$g(f(4)) = g(\underline{\hspace{2cm}}) = \underline{\hspace{2cm}} \quad \text{Notice both came back to } \underline{\hspace{2cm}}, \text{ the number we started with.}$$

**Part 2 — Prove It with Algebra**

**Example 1.**  $f(x) = 3x - 5$  and  $g(x) = (x + 5)/3$

$$f(g(x)) = 3 \cdot \underline{\hspace{2cm}} - 5 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$g(f(x)) = (\underline{\hspace{2cm}} + 5)/3 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

Both give  $x$ , so  $f$  and  $g$  \_\_\_\_\_ inverses.

**You Try 1.**  $f(x) = 2x + 3$  and  $g(x) = (x - 3)/2$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

**You Try 2.**  $f(x) = x + 6$  and  $g(x) = x - 6$

$$f(g(x)) = \underline{\hspace{2cm}} \quad g(f(x)) = \underline{\hspace{2cm}} \quad \text{Inverses? } \underline{\hspace{2cm}}$$

**Part 3 — A Pair That Is NOT Inverse**

Test  $f(x) = 2x + 1$  and  $g(x) = 2x - 1$ . Do not assume - compute.

$$f(g(x)) = 2(\underline{\hspace{2cm}}) + 1 = \underline{\hspace{2cm}} \quad \text{Does this equal } x? \underline{\hspace{2cm}}$$

So they are \_\_\_\_\_ inverses. What is the actual inverse of  $f(x) = 2x + 1$ ?  $f^{-1}(x) = \underline{\hspace{2cm}}$

Lesson: changing a sign is \_\_\_\_\_ the same as undoing the function. You must undo every step, in reverse order.

**Part 4 — Error Analysis**

A student is given  $f(x) = 2(x - 3)^2 + 4$  and claims  $f^{-1}(x) = \sqrt{x - 3} + 4$ .

Test it: is  $f(f^{-1}(x)) = x$ ? \_\_\_\_\_ So the claim is \_\_\_\_\_.

Find the mistake. Undo the steps in reverse: subtract 4, divide by 2, take the square root, then add 3.

$$\text{Correct inverse: } f^{-1}(x) = \underline{\hspace{2cm}}$$

Why does a  $\pm$  show up here? \_\_\_\_\_

What restriction would make the inverse a function? \_\_\_\_\_

## Part 5 — Self-Inverse Functions

A few functions are their own inverse - composing them with themselves gives  $x$ .

1.  $f(x) = x$      $f(f(x)) =$  \_\_\_\_\_ Self-inverse? \_\_\_\_\_

2.  $f(x) = -x$      $f(f(x)) =$  \_\_\_\_\_ Self-inverse? \_\_\_\_\_

3.  $f(x) = 1/x$      $f(f(x)) =$  \_\_\_\_\_ Self-inverse? \_\_\_\_\_

What do their graphs share? \_\_\_\_\_

### Practice

Decide whether each pair is inverse. Show BOTH compositions - one direction is not proof.

1.  $f(x) = 5x$  and  $g(x) = x/5$

\_\_\_\_\_

\_\_\_\_\_

$f(g(x)) =$  \_\_\_\_\_  $g(f(x)) =$  \_\_\_\_\_ Inverses? \_\_\_\_\_

2.  $f(x) = x - 3$  and  $g(x) = 3 - x$

\_\_\_\_\_

\_\_\_\_\_

$f(g(x)) =$  \_\_\_\_\_ Inverses? \_\_\_\_\_ Correct inverse of  $f$ : \_\_\_\_\_

3.  $f(x) = (x + 1)/2$  and  $g(x) = 2x - 1$

\_\_\_\_\_

\_\_\_\_\_

$f(g(x)) =$  \_\_\_\_\_  $g(f(x)) =$  \_\_\_\_\_ Inverses? \_\_\_\_\_

4.  $f(x) = x^2$  and  $g(x) = \sqrt{x}$ . Test with  $x = -4$  as well as  $x = 4$ .

\_\_\_\_\_

\_\_\_\_\_

Does it work for every real  $x$ ? \_\_\_\_\_ Restriction needed: \_\_\_\_\_

5. Write the inverse of  $f(x) = 7x + 2$ , then verify it by composition.

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$f^{-1}(x) =$  \_\_\_\_\_

### Exit Ticket

Decide whether  $f(x) = 4x - 8$  and  $g(x) = (x + 8)/4$  are inverses. Show both compositions.

$f(g(x)) =$  \_\_\_\_\_  $g(f(x)) =$  \_\_\_\_\_ Inverses? \_\_\_\_\_

Why is checking only one direction not enough? \_\_\_\_\_